• 函数求导法则：u, v 可导

(1) $(u \pm v)' = u' \pm v'$

(2) $(u \cdot v)' = u' \cdot v + u \cdot v'$

(3) $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ $(v \neq 0)$

上面n个公式都于 u, v 在某一点 x_0 也是成立的.

(3) 证明：在 $x = x_0$ 点处.

$$\left(\frac{u}{v}\right)'\Big|_{x=x_0} = \lim_{\Delta x \to 0} \frac{\frac{u(x_0+\Delta x)}{v(x_0+\Delta x)} - \frac{u(x_0)}{v(x_0)}}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{u(x_0+\Delta x)\cdot v(x_0) - v(x_0+\Delta x)\cdot u(x_0)}{\Delta x \cdot v(x_0+\Delta x)\cdot v(x_0)}$$

$$= \lim_{\Delta x \to 0} \frac{(u(x_0+\Delta x) - u(x_0))\cdot v(x_0)}{\Delta x \cdot v(x_0+\Delta x)\cdot v(x_0)}$$

$$- \lim_{\Delta x \to 0} \frac{(v(x_0+\Delta x) - v(x_0))\, u(x_0)}{\Delta x \cdot v(x_0+\Delta x)\cdot v(x_0)}$$

$$= \frac{u'(x_0)\cdot v(x_0)}{v^2(x_0)} - \frac{v'(x_0)\cdot u(x_0)}{v^2(x_0)}$$

$$= \frac{u'v - uv'}{v^2}\Bigg|_{x=x_0}$$ ▨

特别地，$\left(\frac{1}{v(x)}\right)' = -\frac{v'(x)}{v^2(x)}$

$(c\cdot u(x))' = c\,u(x)$

（c 为常数）

例：$f(x)=\tan x$ 求 $f'(x)$.

解：$(\tan x)'=\left(\frac{\sin x}{\cos x}\right)'$

$$=\frac{(\sin x)'\cdot\cos x-\sin x\cdot(\cos x)'}{(\cos x)^2}$$

$$=\frac{\cos x\cdot\cos x-\sin x\cdot(-\sin x)}{\cos^2 x}$$

$$=\frac{1}{\cos^2 x}=\sec^2 x$$

故 $(\tan x)'=\sec^2 x=\frac{1}{\cos^2 x}$

例：$f(x)=x\cdot e^x\cdot\ln x+\tan\frac{\pi}{8}+x^2$

求 $f'(x)$.

公式 $(u\cdot v\cdot w)'=u'\cdot v\cdot w+u\cdot v'\cdot w+u\cdot v\cdot w'$

解：$f'(x)=(x\cdot e^x\cdot \ln x)'+(\tan\frac{\pi}{8})'+(x^2)'$

$=x'\cdot e^x\cdot \ln x+x\cdot(e^x)'\cdot \ln x+x\cdot e^x\cdot(\ln x)'$

$+0+2x$

$=1\cdot e^x\cdot \ln x+x\cdot e^x\cdot \ln x+x\cdot e^x\cdot\frac{1}{x}$

$+2x$

$=e^x(\ln x+x\ln x+1)+2x$ ▨

三角函数导数公式：

$(\tan x)'=\sec^2 x=\frac{1}{\cos^2 x}$

$(\cot x)'=-\csc^2 x=-\frac{1}{\sin^2 x}$

$(\sec x)'=\sec x\cdot\tan x=\frac{\sin x}{\cos^2 x}$

$(\csc x)'=-\csc x\cdot\cot x=-\frac{\cos x}{\sin^2 x}$

• 若 $x=\varphi(y)$ 单调可导，$\varphi'(y)\neq 0$，则其反函数 $y=f(x)$ 存在且可导，且 $f'(x)=\dfrac{1}{\varphi'(y)}$.

注：若 $x=\varphi(y_0)$，则 $f'(x_0)=\dfrac{1}{\varphi'(y_0)}$

分析：

$$x \underset{\varphi}{\overset{f}{\rightleftarrows}} y$$

x_0 $\quad$ y_0

$x_0+\Delta x$ $\quad$ $y_0+\Delta y$

$$f'(x_0)=\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x}$$

$$\varphi'(y_0)=\lim_{\Delta y\to 0}\frac{\Delta x}{\Delta y}$$

故 $$f'(x_0)=\frac{1}{\varphi'(y_0)}$$

例：$y=\arcsin x$ 在导

$x\in[-1,1]$ $\quad y\in[-\frac{\pi}{2},\frac{\pi}{2}]$

$y=\arcsin x \quad x=\sin y$

$$\frac{dy}{dx}=\frac{1}{\frac{dx}{dy}}=\frac{1}{\frac{d(\sin y)}{dy}}=\frac{1}{\cos y}$$

即 $y'(x)=\frac{1}{\cos y}$ $(y=\arcsin x)$

由于 $\cos^2 y+\sin^2 y=1$.

$\cos^2 y+x^2=1$

由于 $y\in[-\frac{\pi}{2},\frac{\pi}{2}]$，$\cos y\geq 0$.

故 $\cos y=\sqrt{1-x^2}$.

从而 $y'(x)=\frac{1}{\sqrt{1-x^2}}$ $\quad x\in(-1,1)$

即 $(\arcsin x)'=\frac{1}{\sqrt{1-x^2}}$ ▨

类似地，我们有

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}} \quad x \in (-1,1)$$

- 例 $y = \arctan x$，求导.

解：$x \in (-\infty, +\infty)$ $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$y = \arctan x$，$x = \tan y$

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{(\tan y)'} = \frac{1}{\frac{1}{\cos^2 y}}$$

由于 $\frac{1}{\cos^2 y} = \frac{\sin^2 y + \cos^2 y}{\cos^2 y} = 1 + \tan^2 y$

$= 1 + x^2$.

故 $y'(x) = \frac{1}{1+x^2}$

即 $(\arctan x)' = \frac{1}{1+x^2} \quad x \in (-\infty, +\infty)$

□

类似地 $\boxed{(\operatorname{arccot} x)' = -\frac{1}{1+x^2} \quad x \in (-\infty, +\infty).}$

例：　$f(x)$　　$\varphi(y)$

$y = e^x$，　$x = \ln y$

$x \in (-\infty, +\infty)$，　$y \in (0, +\infty)$

$\frac{dy}{dx} = e^x$　　$\frac{dx}{dy} = \frac{1}{y} = \frac{1}{e^x}$

即 $f'(x) = \frac{1}{\varphi'(y)}$

- 复合函数求导法则（链式法则）

$$\left(f(\varphi(x))\right)' = f'(\varphi(x)) \cdot \varphi'(x).$$

若 $u=\varphi(x)$ 在 x_0 可导，$y=f(u)$ 在 $u_0=\varphi(x_0)$ 可导，其导数为

$$\left.\frac{dy}{dx}\right|_{x=x_0} = f'(u_0)\cdot\varphi'(x_0)$$

分析：

$$\begin{array}{ccccc} x & \xrightarrow{\varphi} & u & \xrightarrow{f} & y \\ x_0 & & u_0 & & y_0 \\ x_0+\Delta x & & u_0+\Delta u & & y_0+\Delta y \end{array}$$

$$\begin{cases} \Delta y \sim f'(u_0)\cdot\Delta u \\ \Delta u \sim \varphi'(x_0)\cdot\Delta x \end{cases}$$ 无穷小等价.

$$\Rightarrow \Delta y \sim f'(u_0)\cdot\varphi'(x_0)\cdot\Delta x$$

从而 $\left(f(\varphi(x))\right)'\Big|_{x=x_0} = f'(\varphi(x_0))\cdot\varphi'(x_0)$

- $f'(\varphi(x)) \neq \left(f(\varphi(x))\right)'$

例 $f(x)=x^3$. $\varphi(x)=2x$

$f(\varphi(x))=(2x)^3=8x^3$.

$\left(f(\varphi(x))\right)'=(8x^3)'=8\cdot 3x^2=24x^2$

但 $f'(x)=3x^2$.

$f'(\varphi(x))=3(\varphi(x))^2=3\cdot(2x)^2=12x^2$

例：求 $y=(2x^2+3x+4)^{50}$ 的导函数.

解：令 $u=2x^2+3x+4$. 则 $y=u^{50}$

$$\frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dx}$$

$$=(u^{50})'_u\cdot(2x^2+3x+4)'_x$$

$$=50u^{49}\cdot(4x+3)$$

$$=50(2x^2+3x+4)^{49}\cdot(4x+3)$$

- 例 $y=\ln|x|$. $(x\neq 0)$. 求 y'.

解：当 $x>0$. $y'=(\ln x)'=\frac{1}{x}$

当 $x<0$ 时. $y'=(\ln(-x))'$ ← $u=-x$

$$=\frac{1}{-x}\cdot(-x)'=-\frac{1}{x}\cdot(-1)=\frac{1}{x}$$

故 $(\ln|x|)'=\frac{1}{x}$ $(x\neq 0)$

• 例 $y=x^{\mu}$ $(x>0,\ \mu$ 为常数)

求导.

解: $y=x^{\mu}=e^{\ln x\cdot\mu}$

$$y'=\left(e^{\underbrace{\ln x\cdot\mu}_{u}}\right)'=e^{\ln x\cdot\mu}\cdot(\ln x\cdot\mu)'$$

$$=e^{\ln x\cdot\mu}\cdot\frac{1}{x}\cdot\mu=x^{\mu}\cdot\frac{1}{x}\cdot\mu$$

$$=\mu x^{\mu-1}$$

故 $$(x^{\mu})'=\mu x^{\mu-1}$$ □

特别地 $$(\sqrt{x})'=\frac{1}{2}\cdot\frac{1}{\sqrt{x}}$$

实际上, 当 $\mu=\frac{1}{3},\frac{1}{5},\frac{1}{7},\cdots\ \frac{2}{3},\frac{4}{3},\cdots$

$x\in\mathbb{R}$ 时, 该式也成立.

如 $(\sqrt[3]{x})' = \frac{1}{3} \cdot \frac{1}{\sqrt[3]{x^2}}$

- 例：$y = \operatorname{arsh} x = \ln(\underbrace{x+\sqrt{x^2+1}}_{u})$.

求 y'.

解：$y' = \frac{1}{x+\sqrt{x^2+1}} \cdot (\underbrace{x+\sqrt{x^2+1}}_{v})'$

$= \frac{1}{x+\sqrt{x^2+1}} \cdot \left(1+\frac{1}{2}\cdot\frac{1}{\sqrt{x^2+1}}\cdot(x^2+1)'\right)$

$= \frac{1}{x+\sqrt{x^2+1}} \cdot \left(1+\frac{1}{2}\cdot\frac{1}{\sqrt{x^2+1}}\cdot 2x\right)$

$= \frac{1}{x+\sqrt{x^2+1}} \cdot \frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}} = \frac{1}{\sqrt{x^2+1}}$

即 $(\operatorname{arsh} x)' = \frac{1}{\sqrt{x^2+1}}$

类似地 $(\operatorname{arch} x)' = \frac{1}{\sqrt{x^2-1}}$

. $\operatorname{arch} x = \ln(x+\sqrt{x^2-1})$

$\operatorname{arsh} x = \ln(x+\sqrt{x^2+1})$.

• 例 $y = e^{\sin^2\frac{1}{x}}$ 求导.

分析: $y = e^u \quad u = \sin^2\cdot\frac{1}{x}$

$u = v^2 \quad v = \sin\frac{1}{x}$

$v = \sin w, \quad w = \frac{1}{x}$

故 $\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dv}\cdot\frac{dv}{dw}\cdot\frac{dw}{dx}$

$= (e^u)'_u \cdot (v^2)'_v \cdot (\sin w)'_w \cdot \left(\frac{1}{x}\right)'_x$

$= e^u \cdot 2v \cdot \cos w \cdot \left(-\frac{1}{x^2}\right)$

$= e^{\sin^2\frac{1}{x}} \cdot 2\sin\frac{1}{x}\cdot\cos\frac{1}{x}\cdot\left(-\frac{1}{x^2}\right)$

上式中可写成 $e^{\sin^2\frac{1}{x}}\cdot\sin\frac{2}{x}\cdot\left(-\frac{1}{x^2}\right)$ □

- (22) $\operatorname{arth} x = \frac{1}{2}\ln\frac{1+x}{1-x}$

$$(\operatorname{arth} x)' = \left(\frac{1}{2}\ln\frac{1+x}{1-x}\right)'$$

$$= \frac{1}{2}\left(\ln(1+x) - \ln(1-x)\right)'$$

$$= \frac{1}{2}\left(\frac{1}{1+x} \cdot - \frac{(-1)}{1-x}\right)$$

$$= \frac{1}{2}\cdot\frac{2}{1-x^2} = \frac{1}{1-x^2}$$

初等函数的导数公式

要背下来！

作业

习题2-2A类

1. (2). (4).

2. (1). (3). (6). (8).

3. (2)

4.

习题2-2B类

1. (5). (6).